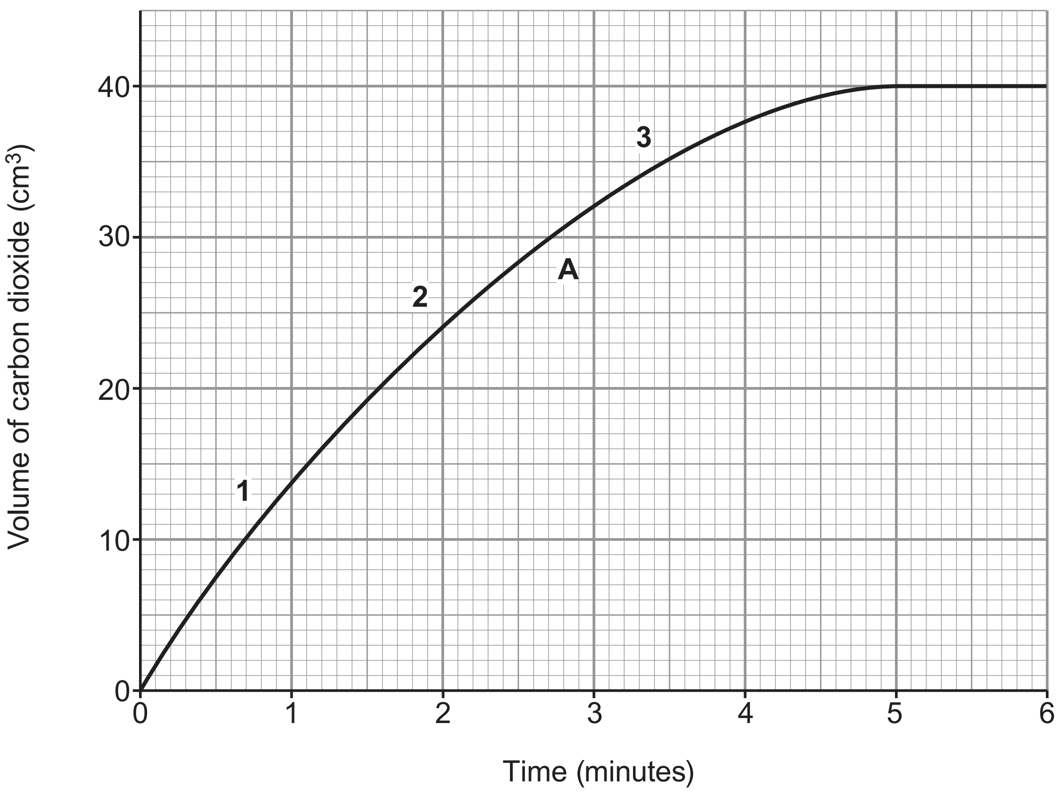
4. Marble chips (calcium carbonate) react with hydrochloric acid to produce carbon dioxide gas as shown in the equation.

$$CaCO_3 + 2HCl \longrightarrow CaCl_2 + CO_2 + H_2O$$

Graph **A** shows the volume of carbon dioxide produced over 6 minutes when the reaction was carried out at 30 °C.



(a) (i) Find the volume of carbon dioxide given off after 2 minutes. [1]

.... cm^3

(ii) Find the time taken for the reaction to finish. [1]

.... minutes

(iii) At which of the following points on graph **A** is the reaction at its fastest?

Tick (✓) the correct box. [1]

1 ☐ **2** ☐ **3** ☐

Examiner only

(iv) After $\frac{1}{2}$ minute, 7.5 cm^3 of carbon dioxide has been produced.

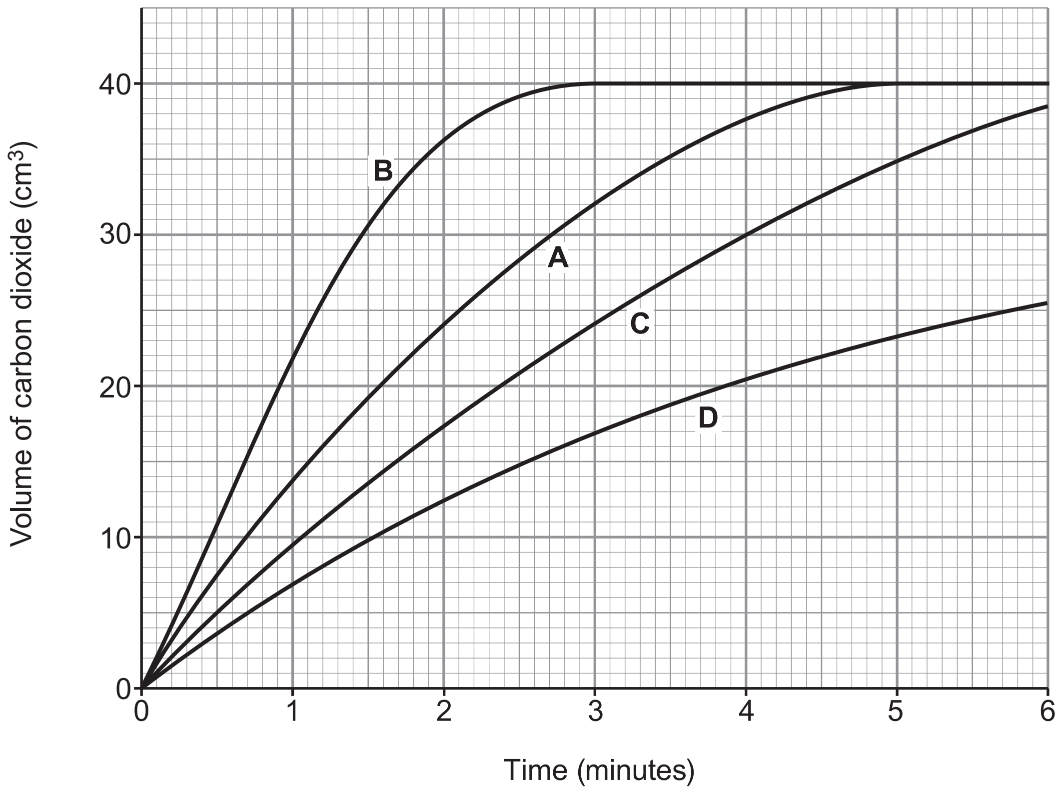
Use the equation to calculate the mean rate of reaction over this time in cm^3/s. [2]

$$\text{mean rate} = \frac{\text{volume of carbon dioxide produced}}{\text{time (in seconds)}}$$

Mean rate = ... cm^3/s

Examiner only

(b) (i) Graphs **B**, **C** and **D** on the grid below show the results for the experiment carried out at **different** temperatures. Graph **A** for the reaction carried out at 30 °C is also shown.



State which of graphs **B**, **C** and **D** shows the results for the experiment carried out at a temperature of 40 °C. [1]

Graph ....

(ii) Using your knowledge of particle theory, underline the correct words in the brackets to complete the following sentence. [2]

At a higher temperature, the particles will have

(more / less / the same amount of) energy, so they will move faster and

there will be **(an equal / a smaller / a greater)** number of successful

collisions per second.

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11

8